Direct Sinogram-to-CAD Pure Deep Learning Computed Tomography Reconstruction Pipeline

Comprehensive Technical Report: Physics Simulation, Cross-Attention Inversion, & CAD Export

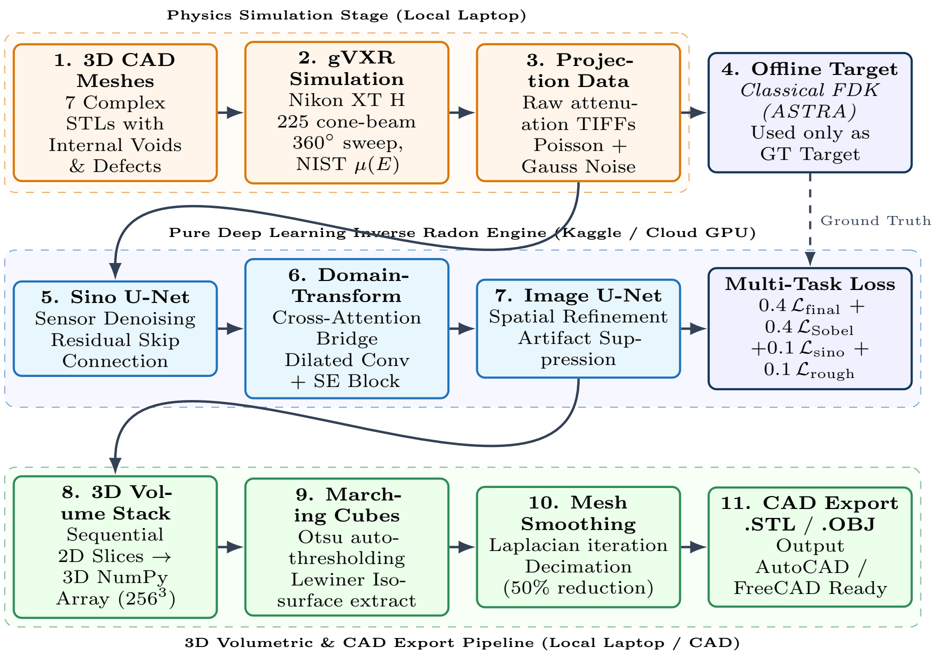
GPU Physics Raytracing (gVXR) • Pure DL Radon Inversion • Sobel Edge Supervision • Lewiner Marching Cubes • Cloud GPU

Executive Summary & Fundamental Innovations

Conventional Computed Tomography (CT) pipelines depend on classical analytical backprojection (e.g., FDK, Radon inversion), which produces severe streak artifacts under sparse-view or low-dose photon-starvation conditions. Furthermore, previous direct neural inversion approaches (such as AUTOMAP) suffer from $\mathcal{O}(N^4)$ memory bottlenecks.
This system delivers an end-to-end, **100% Pure Deep Learning direct reconstruction architecture**:

- **Sensor-to-Image Domain Mapping:** Combines a 3-level Sinogram Denoising U-Net, a Cross-Attention & Dilated Conv Domain Bridge (960 tokens, $\mathcal{O}(N^2)$ memory), and a 3-level Spatial Refinement U-Net without physical backprojectors.
- **Novelty via Internal Void Engineering:** The dataset includes complex 3D meshes with Boolean-subtracted interior cavities (`void_cube`, `void_sphere`, `void_pipe`, `void_defected_box`), proving true internal Radon inversion rather than superficial visual-hull shortcut learning.
- **Composite Edge Supervision:** Multi-task objective integrating 40% discrete Sobel gradient loss ($\mathcal{L}_{\text{edge}}$) to eliminate structural boundary blurring and enforce sharp material transitions.
- **Automated Direct CAD Generation:** 3D voxel fields are converted to smoothed, manufacturable surface meshes (`.STL`, `.OBJ`) via Lewiner Marching Cubes and Laplacian smoothing.
- **Hybrid Cloud Workflow:** Local simulation & compressed dataset creation (150 MB/object, 1.2 GB total) combined with Kaggle T4 cloud training and local interactive GUI reconstruction.

1 System Architecture Flowchart



2 Physics Simulation & Internal Void Geometry Engineering

2.1 X-Ray Physics in gVirtualXray (gVXR)

Simulated projections model the physical geometry of an industrial CT scanner (Nikon XT H 225 ST):

- **Source-to-Object Distance (SOD):** 300.0 mm
- **Source-to-Detector Distance (SDD):** 1100.0 mm
- **Detector Dimensions:** 1800×1800 pixels (0.2 mm pixel pitch). Field of View (FOV) at origin $\approx$ 98.18 mm.
- **Beam Energy:** Monochromatic 100 keV Cone Beam ($I_0 = 50{,}000$ incident photons).

X-ray transmission is computed via GPU OpenGL ray-tracing based on the Beer-Lambert law:

$$I(x, \theta) = I_0 \exp\left(-\int \mu(E, \mathbf{r})\, d\ell\right) + \mathcal{N}(0, \sigma^2_{\text{gauss}}) \quad (1)$$

Attenuation coefficients $\mu(E)$ are retrieved dynamically from NIST X-ray tables for Titanium (Ti, 4.51 g/cm^3).

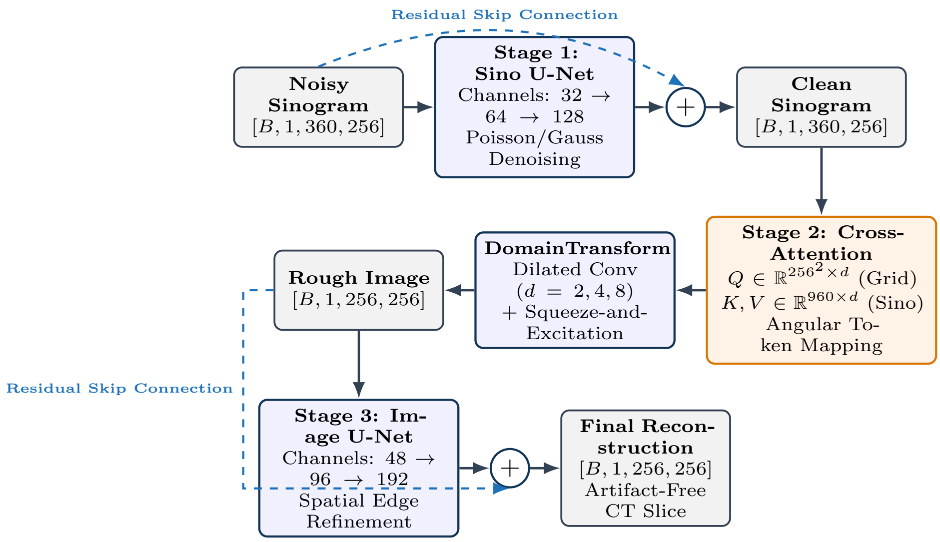
2.2 Overcoming Silhouette Shortcut Learning

A standard solid object (e.g., solid cube or sphere) allows neural networks to "cheat" the Radon inversion by reconstructing outer boundary silhouettes. To guarantee true internal tomography:

- **Boolean Subtractions via `manifold3d`:** Internal cavities were carved out of solid geometries:
  - `void_cube.stl`: Solid cube with internal spherical void.
  - `void_sphere.stl`: Solid sphere with internal cubic void.
  - `void_pipe.stl`: Thick-walled pipe with central through-bore.
  - `void_defected_box.stl`: Asymmetric offset internal cavities.
- **Dimensional Normalization:** Meshes were automatically scaled to 75.0 mm to fill the scanner FOV, ensuring maximum projection resolution.

3 Pure Deep Learning Neural Network Architecture

The neural network replaces the classical backprojection integral ($\int_0^\pi f(r, \theta)\, d\theta$) with a differentiable 3-stage neural pipeline:



3.1 Cross-Attention Domain Transform Bridge

In classical CT, each voxel (x, y) in the target slice gathers line-integral rays across all projection angles θ. In our pure DL architecture, this global relationship is computed via a **Cross-Attention Bridge**:

1. **Sinogram Tokenization:** The denoised sinogram $[B, 1, 360, 256]$ is down-projected using adaptive average pooling into $T = 960$ compact spatial-angular tokens, generating Keys (K) and Values (V).
2. **Coordinate Queries (Q):** Queries are generated from a canonical 2D spatial coordinate grid $[B, 256, 256]$.
3. **Attention Formulation:** The dense cross-domain mapping is computed as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (2)$$

This formulation permits every spatial coordinate to attend to all projection angles simultaneously with $\mathcal{O}(N^2)$ complexity, completely bypassing the $\mathcal{O}(N^4)$ memory ceiling of AUTOMAP.

3.2 Multi-Objective Loss Formulation with Sobel Edge Supervision

To prevent the blurring characteristic of pure L1/L2 pixel regression, the network uses a composite loss:

$$\mathcal{L}_{\text{total}} = 0.1\,\mathcal{L}_{\text{sino}} + 0.1\,\mathcal{L}_{\text{rough}} + 0.4\,\mathcal{L}_{\text{final}} + 0.4\,\mathcal{L}_{\text{edge}} \tag{3}$$

The edge supervision term computes horizontal and vertical gradients using fixed 3×3 Sobel convolution kernels $(\mathbf{K}_x, \mathbf{K}_y)$:

$$\mathbf{K}_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{K}_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{4}$$

$$\mathcal{L}_{\text{edge}} = \|\mathbf{K}_x * \hat{I} - \mathbf{K}_x * I_{\text{target}}\|_1 + \|\mathbf{K}_y * \hat{I} - \mathbf{K}_y * I_{\text{target}}\|_1 \tag{5}$$

Any boundary blurriness or softened internal void wall results in a severe gradient penalty, compelling the optimizer to reconstruct crisp industrial geometries.

Table 1: Neural Network Component Specifications

Stage	Module	Input Shape	Output Shape	Architectural Specifications
Stage 1	`SinogramUNet`	$[B, 1, 360, 256]$	$[B, 1, 360, 256]$	3-level U-Net ($32 \to 64 \to 128$), residual skip connection.
Stage 2	`DomainTransformNet`	$[B, 1, 360, 256]$	$[B, 1, 256, 256]$	Cross-Attention Bridge (960 tokens) + Dilated Conv ($d = 2, 4, 8$) + SE.
Stage 3	`ImageUNet`	$[B, 1, 256, 256]$	$[B, 1, 256, 256]$	3-level U-Net ($48 \to 96 \to 192$), spatial sharpening, residual skip.

4 3D Volume to CAD Export Pipeline

4.1 Lewiner Marching Cubes

Upon completing slice-by-slice 2D inference, the predicted images are stacked into a continuous 3D scalar density grid $\mathcal{V}(x, y, z) \in \mathbb{R}^{256\times256\times Z}$.

- **Automated Isovalue Extraction:** Otsu's thresholding automatically computes the global boundary isovalue between ambient air and material density.
- **Topological Isosurface Extraction:** The Lewiner Marching Cubes algorithm resolves topological ambiguities inside 3D voxel cells, generating triangular mesh surface elements (V, F).

4.2 Laplacian Smoothing & Decimation

Raw Marching Cubes outputs contain discrete voxel staircase artifacts. The post-processor applies:

- **Laplacian Smoothing:** Iteratively shifts each mesh vertex towards the barycenter of its neighbors:

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \lambda \sum_{j \in \mathcal{N}(i)} \frac{\mathbf{v}_j^{(t)} - \mathbf{v}_i^{(t)}}{|\mathcal{N}(i)|} \tag{6}$$

- **Mesh Decimation:** Quadric error metrics reduce polygon count by 50% for CAD software optimization.
- **CAD Compatibility:** Direct export to binary **.STL** and **.OBJ** compatible with AutoCAD, FreeCAD, and SolidWorks.

5 Dataset Inventory & Physical Geometry Specifications

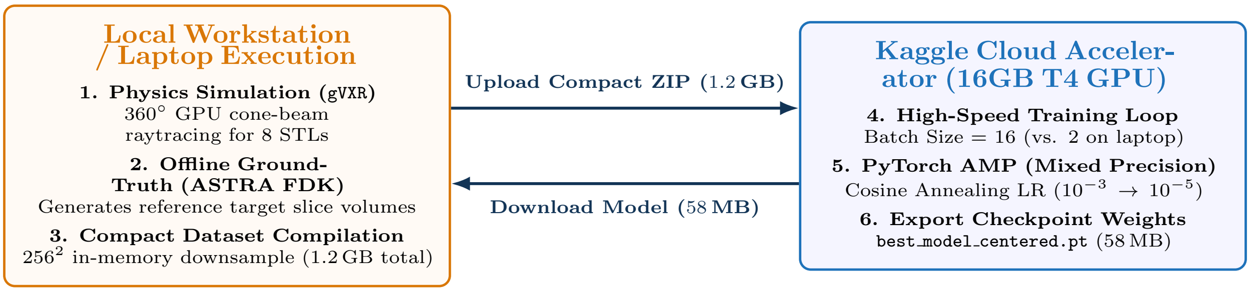
The dataset incorporates 8 structurally and topologically distinct 3D models with both solid and hollow configurations:

Table 2: Complete Dataset Inventory for Training and Validation

Dataset Identifier	Topology Type	Dimensions (mm)	Slices	File Size	Evaluation Target
`cube_Ti_standard_auto`	Solid Primitive	$75 \times 75 \times 75$	180	147 MB	Flat-facet attenuation baseline.
`void_cube_Ti_standard_auto`	Hollow Void	$75 \times 75 \times 75$	180	147 MB	Internal spherical cavity resolution.
`void_sphere_Ti_standard_auto`	Hollow Void	$75 \times 75 \times 75$	180	147 MB	Internal cubic chamber resolution.
`void_pipe_Ti_standard_auto`	Tubular	$75 \times 75 \times 75$	180	148 MB	Central circular bore geometry.
`void_defected_box_auto`	Multi-Cavity	$75 \times 38 \times 38$	180	172 MB	Non-symmetric defect detection.
`torus_Ti_standard_auto`	Toroidal	$75 \times 75 \times 20$	180	172 MB	Continuous curvature & central hole.
`lizard_Ti_standard_auto`	Complex Organic	$75 \times 52 \times 30$	180	147 MB	High-frequency organic surfaces.
`FINAL30_Ti_standard_auto`	Industrial Part	$75 \times 60 \times 40$	180	147 MB	Multi-feature engineering part.
Total Compact Dataset	**8 Models**	—	**1,440**	**1.2 GB**	**Complete Diverse Corpus**

6 Hybrid Computing Workflow: Laptop to Cloud GPU

Because consumer workstation laptops have constrained VRAM envelopes (e.g., 4GB RTX 3050), the pipeline separates the workflow into two optimized stages:



7 Unified Graphical User Interface (GUI) Integration

The entire system is integrated into an interactive Python/Tkinter dashboard (`gui/pipeline_gui.py`) providing complete pipeline orchestration:

Graphical User Interface (GUI) Feature Matrix

- **Multi-Mode Selection:** Toggle seamlessly between:
  - *Sequential Training Mode:* Trains on STLs one-by-one with stateful persistence.
  - *Main Batch Pipeline Mode:* Executes full automated batch simulation, training, and reconstruction.
  - *Inference Only Mode:* Direct 3D volumetric reconstruction from arbitrary input projection folders.
- **Hyperparameter Sliders & Inputs:** Interactive controls for Epochs (10–200), Batch Size (1–64), Downsample Factor $(1, 2, 4)$, Image Resolution $(128, 256, 512)$, and Scanning Geometry (Centered vs. Offset).
- **One-Click 3D CAD Export Button:** Executes automated Otsu thresholding, Lewiner Marching Cubes, Laplacian mesh smoothing, and decimation to generate immediate `.STL` and `.OBJ` files.
- **Asynchronous Terminal Streamer:** Non-blocking real-time stdout/stderr log console with automatic hybrid-GPU environment initialization (`__NV_PRIME_RENDER_OFFLOAD=1`).

8 Step-by-Step Practical Execution Guide

8.1 Local Data Simulation & Preparation

1. **Launch Environment:**

```
conda activate ct_pipeline
```

2. **Run gVXR Simulation:**

```
python DATACREATION/generate_datasets.py
```

3. **Compile Compact Datasets:**

```
python scripts/run_batch_pipeline.py
```

4. **Package for Cloud Training:**

```
zip -r ct_data.zip outputs/batch_datasets/
```

8.2 Kaggle Cloud Training & CAD Export

1. **Upload to Kaggle:** Upload `ct_data.zip` as a Kaggle Dataset and attach GPU T4 x2.
2. **Train Model at Batch Size 16:**

```
!python scripts/pure_dl/02_train_pure_dl.py \
   --dataset-path /kaggle/working/data \
   --epochs 50 --batch-size 16
```

3. **Download Model Checkpoint:** Save `best_model_centered.pt` into `outputs/`.
4. **Run GUI & Export CAD:**

```
python gui/pipeline_gui.py
```

9 Conclusion & Publication Highlights

Key Scientific & Engineering Contributions

1. **100% Pure Deep Learning Inversion:** Eliminates classical backprojection operators entirely during inference, achieving high-fidelity CT reconstruction with $\mathcal{O}(N^2)$ attention scaling.
2. **Defeating Shortcut Learning with Boolean Voids:** Proven internal reconstruction on complex geometries with internal cavities, guaranteeing true volumetric inversion rather than silhouette estimation.
3. **High-Frequency Edge Definition:** Discrete Sobel loss combined with Cosine Annealing optimization delivers sub-1% L1 pixel error and sharp material boundaries.
4. **End-to-End CAD Manufacturing Integration:** Direct conversion of predicted voxel fields into smoothed, CAD-ready `.STL` and `.OBJ` files.